

The **HAF-EPA** (Hybrid AI Framework for Employee Project Allocation) pipeline was executed to train, validate, and evaluate a hybrid employee recommendation model.

The framework analyzes employee skills, experience, availability, project history, soft-skill compatibility, and task-specific requirements to identify the most suitable employees for project assignments.

The following sections summarize the key stages involved in the model training, evaluation, and recommendation generation process.

1

STEP 1: Preparing Labeled Dataset

--- STEP 1: Prepare my labeled dataset ---

 Mark as Completed

2

STEP 2: Split: Training / Test Dataset (80/20)

=== STEP 2: Split: Training / Test dataset (80/20) ===

Train shape:	(1800000, 71)
Test shape:	(450000, 71)

 Status: Completed

3 STEP 3: Balance Training Dataset



=== STEP 3: Balance Training dataset ===

Referenced shape:

(138818, 71)

✓ Status: Completed

4 STEP 4: Start Train HAF-EPA Model



=== STEP 4: Start train HAF-EPA model ===

Model saved location:

/Users/mcc-hui/Documents/Thesis/HAF-EPA/output/HAF-EPA.joblib

✓ Status: Completed

5 STEP 5: Validate Test Dataset (20%)



=== STEP 5: Start validate test dataset(20%) ===

Accuracy:

0.8839797777777778

✓ Status: Completed

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STEP 6: Generate Hybrid Recommendations



=== STEP 6: Generate hybrid recommendations on test dataset ===

Top 10 Recommendations:

	employee_id	project_id	predicted_label	task_context_match_score	soft_skill_compatibility_score	recommendation_reason
0	E00294	P00313	1	0.1	6.804673	matched required skills: 4; weighted skill match.
1	E00294	P00518	1	0.1	6.804673	matched required skills: 4; weighted skill match.
2	E00294	P00705	1	0.1	6.804673	matched required skills: 4; weighted skill match.
3	E00294	P00295	1	0.1	6.804673	matched required skills: 4; weighted skill match.
4	E00294	P00461	1	0.1	6.804673	matched required skills: 4; weighted skill match.
5	E00294	P00556	1	0.1	6.804673	matched required skills: 4; weighted skill match.
6	E00294	P00070	1	0.1	6.804673	matched required skills: 4; weighted skill match.
7	E00294	P00621	1	0.1	6.804673	matched required skills: 4; weighted skill match.
8	E00184	P00443	1	0.1	6.848421	matched required skills: 4; weighted skill match.
9	E00184	P00404	1	0.1	6.848421	matched required skills: 4; weighted skill match.

Showing top 10 rows × 16 columns

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STEP 7: Create CSV and Excel Files



=== STEP 7: Create CSV and Excel files ===



Feature importance saved to:

/Users/firoz-fau/Documents/Thesis/HAF-EPA/output/
HAF-EPA_feature_importance.csv



Balanced training data saved to:

/Users/firoz-fau/Documents/Thesis/HAF-EPA/output/
HAF-EPA_balanced_training_dataset.csv



Test predictions CSV saved to:

/Users/firoz-fau/Documents/Thesis/HAF-EPA/output/
HAF-EPA_test_predictions.csv



Recommendation Excel saved to:

/Users/firoz-fau/Documents/Thesis/HAF-EPA/output/
HAF-EPA_test_recommendations.xlsx



Evaluation report saved to:

/Users/firoz-fau/Documents/Thesis/HAF-EPA/output/
HAF-EPA_model_evaluation.txt



Status: Completed

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STEP 8: Create Top 200 Employee Summary Excel



=== STEP 8: Create top 200 employee summary Excel sheet ===



Excel saved to:

/Users/firoz-fau/Documents/Thesis/HAF-EPA/output/
HAF-EPA_top_200_employees.xlsx



CSV saved to:

/Users/firoz-fau/Documents/Thesis/HAF-EPA/output/
HAF-EPA_employee_reference.csv



Status: Completed

10 Top 10 Important Features



=== TOP 10 Important features ===

	Feature	Importance
1	avg_past_performance_score	0.379548
2	soft_skill_compatibility_score	0.310000
3	task_context_match_score	0.141254
4	weighted_skill_match_score	0.010839
5	skill_match_score	0.017049
6	employee_skill_coverage	0.015251
7	avg_experience_on_required_skills	0.013686
8	related_skill_match_score	0.012492
9	matched_skill_count	0.012245
10	availability_fit_score	0.010435

Top 10 Employees Details

=== TOP 10 Employees details ===

	employee_id	full_name	department	has_soft_skill_score	avg_allocation_percent	ma_compatibility_score
1	E00006	Carrie Lawrence	Operations	6.659634	40.0	6.659634
2	E00119	Lisa Dixon	Product	6.865152	30.0	6.865152
3	E00071	Jose Henderson	IT Support	6.788889	20.0	6.788889
4	E00015	Bobby Hall	IT Support	6.838158	50.0	6.838158
5	E00195	Robert Peterson	Software Engineering	6.603896	80.0	6.603896
6	E00091	Brian Rodriguez	Data Science	6.903191	30.0	6.903191
7	E00079	Thomas Bright	Product	6.593023	80.0	6.593023
8	E00083	Elijah Stewart	IT Support	6.888043	20.0	6.888043
9	E00063	Randy Lee	Software Engineering	6.615115	70.0	6.615115
10	E00261	Diane Evans	Data Science	6.767059	30.0	6.767059

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10 STEP 10: Final Summary and Completion



=== STEP 10: Final summary and completion ===



Status: All tasks completed successfully!